Assignment: Quantitative Evaluation

Why do I do this?

There is a big difference between learning to run an experiment and actually running an experiment. This assignment is designed to give you the experience of preparing an experiment, meeting participants and interacting with them, and collecting and analyzing results from the experiment. This will be helpful for you to plan the course project and, hopefully, your other research projects.

What do I do?

You will conduct an experiment that tests pointing performances using WebFit (https://adildsw.github.io/WebFitt/). You can choose any two independent variables (e.g., hand, input device, posture, etc.).

The study should follow the common procedure of running a user study. It means that you should greet the participant, get demographic information, explain the tasks, moderate between blocks and conditions, and say thank you after completion of the study.

The WebFit should work on Mac, Windows, and Linux, as long as you are using a modern web browser (e.g., Chrome, Safari, Opera). Every time the participant finishes a session, the webpage will automatically initiate the downloading of a zip file containing log files. The three files in the zip file are CSV (comma-separated values) files, so if you change the file extension to .csv, you'll be able to open it on most spreadsheet programs (e.g., Microsoft Excel, Google Sheets) and statistics programs (e.g., SPSS, SAS JMP, R).

After running the experiment, **you will analyze the results (Task Completion Time and Error Rate),** including running a hypothesis test. For hypothesis testing, you may use any tools of your choice, including R, Matlab, SPSS, Python, and GoStats (http://www.yorku.ca/mack/GoStats/). GoStats may be easy to use once you get used to the data format, which is explained in their web documentation. Or, you can use R or Python for analysis. Example Python code for analyzing in-class study data is on Blackboard.

There should be at least **4 participants**. You may recruit anyone, including those in the class, but you can't be a participant in your own study.

Who do I work with?

Conducting an experiment is not easy, as there are several things to prepare and control. So for this assignment, you can work with one more person as a team. If you already have someone you want to work with, then you can work with them. If not, we'll find a partner for you.

Submit your preference using this: https://forms.gle/Bt38oNe8xwCBLFDAA

Your report

Each team submits one report. The report should include the following sections:

- **Names** and **student IDs**
- **Study Design:**
  - Explain the study design. Within/between-subjects design, independent and dependent variables, trials and blocks used, study setting, and tasks.
  - Explain how the participants were recruited, who they were (demographics), and how long it took for each participant to complete the study (on average)
- **Study Results:**
  - Descriptive statistics. Show the results using graphs visualizing average completion time and error across different blocks for each method and explain the numbers in the paragraph.
  - Hypothesis testing results (aka stats)
  - Briefly discuss the results in your words.
- **Reflection:**
  - Describe your reflection in a short paragraph. E.g., what you've learned, challenges you've faced, and/or what you liked or disliked about this study.

You may find Jacob O. Wobbrock's guide to writing CHI papers useful in writing experiment design/results and also your final project report (http://faculty.washington.edu/wobbrock/pubs/Wobbrock-2015.pdf).

Grading (15 points)

- **Completeness (9 points):** Cover all required components listed above in your report.
- **Depth (3 points):** Include thoughtful analysis and reflections.
- **Clarity (3 points):** The writing should be clear to understand, and the visual materials in the report should effectively communicate the idea.
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Academic Honesty

It is okay if the results are not significant or pretty. With only four participants, the results may not be that different. Or maybe those two input devices are not very different. But do not ever attempt to modify or create the results, or copy someone else's report.

How do I submit?

Make a single PDF file of your report and submit it to Blackboard.

Late policy

Deadline is Apr 29 (Wed) 23:59pm. You may submit the report up to two days after the deadline, with 20% and 50% penalty.

Questions?

If you have any questions about the assignment, feel free to send Slack messages or email to the instructor.